

L1.5 Differentiation Formulae

§2.3 — what you need to know

The five rules

① **constant** $\frac{d}{dx}(c) = 0$

② **power** $\frac{d}{dx}(x^n) = nx^{n-1}$ every real n — negative and fractional too

③ **sum** $\frac{d}{dx}(f \pm g) = \frac{df}{dx} \pm \frac{dg}{dx}$ $\frac{d}{dx}(cf) = c \frac{df}{dx}$

THE TWO THAT ARE NOT TERM BY TERM

④ **product** $\frac{d}{dx}(f(x)g(x)) = f(x)g'(x) + g(x)f'(x)$

⑤ **quotient** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{gf' - fg'}{g^2}$

“low d high, minus high d low, over the square of what’s below”

Each function is **kept once and differentiated once**. The quotient’s numerator is a subtraction, so its order is not free — swapping the two products gives the negative of the right answer.

REWRITE AS A POWER FIRST

The power rule applies to x^n and nothing else, so roots and reciprocals are rewritten *before* they are differentiated. **This is the step that gets skipped.**

$$\sqrt{x} \rightarrow x^{1/2} \quad \frac{1}{x^3} \rightarrow x^{-3} \quad \sqrt[3]{x^2} \rightarrow x^{2/3}$$

And you can **always take out the c** : $\frac{d}{dx}(20x^{1/2}) = 20 \frac{d}{dx}(x^{1/2}) = 20 \cdot \frac{1}{2}x^{-1/2} = \frac{10}{\sqrt{x}}$

WHERE nx^{n-1} COMES FROM

Straight out of the definition. The coefficients of $(x+h)^3$ are row 3 of **Pascal’s triangle** — 1 3 3 1 — and every term then has one of **three fates**:

$$\frac{(x+h)^3 - x^3}{h} = \frac{\overbrace{x^3 - x^3}^{\text{cancels}} + \overbrace{3x^2h}^{h \text{ divides out}} + \overbrace{3xh^2 + h^3}^{\rightarrow 0}}{h} = 3x^2 + 3xh + h^2 \xrightarrow{h \rightarrow 0} 3x^2$$

The coefficient is the second entry of row n , which is always n ; the exponent is always one less. That is the whole rule — and it is why it works for every n .

SECOND DERIVATIVES AND HORIZONTAL TANGENTS

$f''(x) = \frac{d^2y}{dx^2}$ is the same five rules **run twice**. Nothing new.

A **horizontal tangent** is a slope of 0, so you find one by **solving** $f'(x) = 0$. For $y = x^3 - x$:
 $3x^2 - 1 = 0 \Rightarrow \boxed{x = \pm \frac{1}{\sqrt{3}}}$. *This is the first time the derivative is used to **find** something instead of to report it.*

M (max or min) on $f \rightarrow$ a **zero** on $f' \rightarrow$ —

IP on $f \rightarrow$ an M on $f' \rightarrow$ a **zero** on f''

WATCH OUT

~~✗ the derivative of a product is the product of the derivatives $\rightarrow$ it is $fg' + gf'$ — expanding $(x^2 - 1)(x + 4)$ proves it~~

~~✗ the quotient's numerator is $fg' - gf'$ $\rightarrow$ it is $gf' - fg'$ — low d high *minus* high d low~~

~~✗ $\sqrt{x}$ needs its own rule $\rightarrow$ rewrite it as $x^{1/2}$ and it *is* the power rule~~

~~✗ f'' needs a new technique $\rightarrow$ it is the same rules, run twice~~

~~✗ the power rule works on 2^x $\rightarrow$ the *exponent* has to be the constant, not the base~~

~~✗ every quotient needs the quotient rule $\rightarrow$ if the bottom is a single power, split the fraction instead~~